

Changyi Li

Doctoral Researcher · School of Electrical Engineering (ELEC), Aalto University · advised by Yu Xiao

Doctoral researcher in computer vision, working on visual anomaly detection and cross-domain action recognition.

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RESEARCH INTERESTS

Visual anomaly detection (industrial & video) · cross-domain few-shot action recognition · multimodal sensing · large language models for structured data

EDUCATION & EXPERIENCE

- 2023 – now **Ph.D. Candidate, Computer Vision**
Aalto University, School of Electrical Engineering (ELEC) — Espoo, Finland
- 2022 – 2023 **Research Engineer**
Huawei, Chengdu Research Institute — Chengdu, China
- 2019 – 2022 **M.Sc.**
University of Electronic Science and Technology of China (UESTC) — Chengdu, China

HONORS & AWARDS

- 2026 **2nd place — Smart Vending Cabinet Product-Transaction Video Recognition Challenge**
[PRCV 2026 Challenge](#)
- 2026 **3rd place — Track 1 (Micro-Action Recognition); 2nd place — Track 2 (Multi-label Micro-Action Detection)**
[3rd Micro-Action Analysis Grand Challenge \(MAC 2026\)](#), in conjunction with ACM Multimedia 2026

PUBLICATIONS

PUBLISHED & ACCEPTED

- [1] **UniSLAD: A Unified Framework for Structural and Logical Industrial Visual Anomaly Detection**
Changyi Li, Chao Yang, Yu Xiao, Kari Tammi
[IEEE CASE](#) 2026
- [2] **Adaptive Multi-Granularity Temporal Modeling for Weakly Supervised Video Anomaly Detection**
Changyi Li, Yu Xiao
[PRCV](#) 2026
- [3] **Training-Free Logical and Structural Anomaly Detection via Calibrated Fusion**
Changyi Li, Miao Yu, Kai Dong, Yu Xiao
[PRCV](#) 2026
- [4] **Reachability Is Not Generalization: Understanding Verb–Noun Decomposition in Assembly Action Recognition**
Changyi Li, Yu Xiao
[BMVC](#) 2026
- [5] **Multi-Modal Sensing-Assisted Cross-Environment Beam Prediction Towards 6G**
Lei Wang, Changyi Li, Kai Dong, Sergiy A. Vorobyov, Esa Ollila
[IEEE PIMRC](#) 2026 · [Oral](#)

- [6] **A Portable Hybrid Quantum–Classical Neural Network for Continual Learning**
Zeyu Yang, Yunpeng Bai, **Changyi Li**, Yu Xiao
IEEE QCN 2026
- [7] **Exploring Human–AI Collaboration in E-Textile Design: A Case Study on Flex Sensor Placement for Shoulder Motion Detection**
Zhuchenyang Liu, Yao Zhang, Yalan He, Hilla Paasio, **Changyi Li**, Guna Semjonova, Yu Xiao
Proc. ACM Hum.-Comput. Interact. 2026
- [8] **Enhancing Knowledge Graph Interactions: A Comprehensive Text-to-Cypher Pipeline with Large Language Models**
Chao Yang, **Changyi Li**, Xiaodu Hu, Hao Yu, Jinzhi Lu
Information Processing & Management 2025 · **Journal**
- [9] **Revisiting the Impact of Domain Similarity on the Performance of Cross-Domain Few-Shot Activity Recognition**
Changyi Li, Yu Xiao
IEEE BigData 2025
- [10] **S2TKD: Dual-Student Knowledge Distillation for Industrial Visual Anomaly Detection and Localization**
Changyi Li, Chao Yang, Yu Xiao
IEEE BigData 2025 · **Oral**
- PREPRINTS & UNDER REVIEW
- [11] **Pre-Decoding Acoustic Triage for Budgeted Vision-Language Captioning of Untrimmed Egocentric Video**
Masoud Jalayer, **Changyi Li**, Yu Xiao
IEEE BigData 2026 · under review
- [12] **OffRoad-OOD: Benchmarking and Terrain-Conditional Calibration for Off-Road OOD Segmentation**
Changyi Li, Yu Xiao
ICRA 2027 · under review
- [13] **Probing Representation Sensitivity in Time-Series Anomaly Detection Benchmarks with Spectral Whitening**
Changyi Li, Yu Xiao
ICASSP 2027 · under review
- [14] **Closing the Estimation Gap: Training-Free Transductive Inference for Cross-Domain Few-Shot Action Recognition**
Changyi Li, Zeyu Yang, Yu Xiao
WILD VISION Workshop, BMVC 2026 · under review
- [15] **Federated Tucker LLM Fine-Tuning under Multi-Mode Rotational Ambiguity for Beam Prediction**
Kai Dong, **Changyi Li**, Lei Wang, Sergiy A. Vorobyov, Stefan Werner
ICASSP 2027 · under review
- [16] **Uncertainty Quantification for LLM-Empowered Channel Prediction via Online Conformal Prediction**
Lei Wang, Kai Dong, **Changyi Li**, Sergiy A. Vorobyov, Esa Ollila
ICASSP 2027 · under review
- [17] **Vision-Aided LLM-Empowered Online Conformal Beam Prediction**
Lei Wang, Kai Dong, **Changyi Li**, Sergiy A. Vorobyov, Esa Ollila
IEEE Wireless Communications Letters · under review

- [18] **Compressed LLM Preprogramming for Vision-Aided Beam Prediction in Vehicular Networks**
Kai Dong, Lei Wang, **Changyi Li**, Sergiy A. Vorobyov, Stefan Werner
IEEE Wireless Communications Letters · under review
- [19] **Large Language Model-Empowered Beam Prediction: Methods and Future Directions**
Changyi Li, Lei Wang, Kai Dong, Sergiy A. Vorobyov, Robert W. Heath Jr.
IEEE Communications Magazine · under review